

Algorithms: Assignment sheet 2

Due date: October 11, 2026

1. (a) Suppose you are given a directed graph $G = (V, E)$ and two vertices $s, t \in V$. Give a polynomial-time algorithm to find the maximum number of internally vertex-disjoint s - t paths. Two s - t paths P_1 and P_2 are said to be internally vertex-disjoint if $V(P_1) \cap V(P_2) = \{s, t\}$.
(b) State and prove the corresponding min-max theorem for internally vertex-disjoint s - t paths, analogous to the max-flow min-cut theorem.
2. (a) Let $G = (V, E)$ be an undirected graph. Give a polynomial-time algorithm to compute a minimum vertex cut in G . A vertex cut is a set $S \subseteq V$ such that removing S disconnects G into at least two nonempty components.
(b) Can this problem be solved using a Karger-style contraction algorithm? Give a brief argument supporting your answer.
3. Let $G = (A \cup B, E)$ be a bipartite graph with bipartition (A, B) , where $|A| = |B|$. A matching is a set of edges no two of which share an endpoint. A matching is called *perfect* if it has size $|A|$.
Prove that G has a perfect matching if and only if Hall's condition holds: $|N(S)| \geq |S|$ for every $S \subseteq A$, where $N(S) = \{v \in B : (u, v) \in E \text{ for some } u \in S\}$.
4. Let $G = (V, E)$ be an undirected graph. For any $X \subseteq V$, let $\delta(X)$ denote the number of edges in the cut $(X, V \setminus X)$; that is, the number of edges with one endpoint in X and the other in $V \setminus X$.
For any two subsets $A, B \subseteq V$, prove that $\delta(A) + \delta(B) \geq \delta(A \cap B) + \delta(A \cup B)$.
5. Prove the following statement about minimum cuts in an undirected graph $G = (V, E)$. Suppose $(S, V \setminus S)$ is a minimum s - t cut, and let $u, v \in S$. Show that there exists a minimum u - v cut $(U, V \setminus U)$ such that $U \subseteq S$ or $V \setminus U \subseteq S$.
6. Let $G = (V, E)$ be an undirected graph. For any nonempty subset $S \subseteq V$, let $E(S)$ denote the set of edges with both endpoints in S . Define the density of S as $\rho(S) = \frac{|E(S)|}{|S|}$. Give a polynomial-time algorithm to find a subset $S \subseteq V$ of maximum density.
7. We want to find a global min-cut $(A, V \setminus A)$ in an undirected graph $G = (V, E)$ (with unit edge weights and $|V| = n$) such that $|A|$ is the smallest among all global min-cuts in G . Show a randomized polynomial time algorithm with running time $O(n^2 \cdot \text{poly log } n)$ that finds such a global min-cut in G with probability at least $3/4$.
8. Suppose that at each step of the simple contraction algorithm for min-cut, instead of choosing a random edge for contraction, we choose two vertices at random and merge the two vertices into a single vertex. Show that there are inputs on which the probability that this modified algorithm finds a min-cut is exponentially small.

9. In the algorithm Preflow-Push with FIFO we saw that the number of relabels is $O(n^2)$, the number of saturating pushes is $O(mn)$, and the number of non-saturating pushes is $O(n^3)$. Show that the entire algorithm can be implemented to run in $O(n^3)$ time. Recall that n and m are the number of vertices and edges in G , respectively.
10. Consider the following network - the numbers next to the edges denote their capacities and the capacity r of the edge (a, b) is $(\sqrt{5} - 1)/2$.

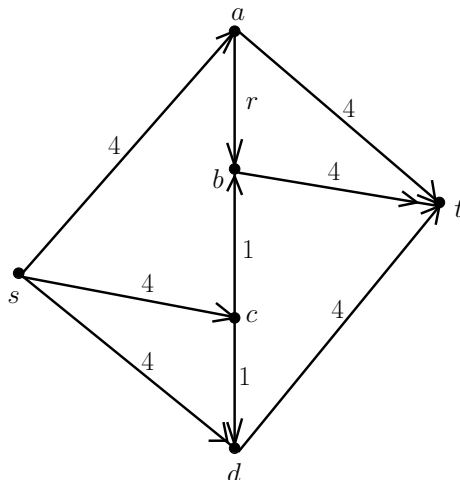


Figure 1: An example where Ford-Fulkerson algorithm does not terminate.

Let the initial flow be along the path $p_0 = (s, c, b, t)$. Take $p_1 = (s, a, b, c, d, t)$, $p_2 = (s, c, b, a, t)$, $p_3 = (s, d, c, b, t)$.

Show that $p_0(p_1, p_2, p_1, p_3)^*$ is an infinite sequence of s - t paths along which positive flow can be sent. The flow value does not converge to the maximum value 9.